Name: ______________________________ Tutorial group: ________________

Matriculation number: | | | | | | | | | |

6 November 2023 **MH1812 Test 2** 60 minutes

For graders only	Question	1	2	3(a)	3(b)	4	Bonus	Total
	Marks							

QUESTION 1. **(20 marks)**

Solve the following linear recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2}, \quad n \geq 2$$

with initial values $a_0 = 2$ and $a_1 = 9$.

Solution: The characteristic equation of the linear recurrence relation is:

$$x^2 - 6x + 9 = 0$$

It leads to two same solution $s = 3$, hence

$$a_n = u \cdot 3^n + v \cdot n \cdot 3^n$$

Substituting the initial values, get

$$a_0 = u = 2 \text{ and } a_1 = 3u + 3v = 9$$

Solving above, we obtain $u = 2$ and $v = 1$, hence $a_n = 2 \cdot 3^n + n \cdot 3^n = (2 + n) \cdot 3^n$.

QUESTION 2. **(15 marks)**

Prove by membership table that for any sets A, B, and C,

$$A - (B \cup C) = (A - B) \cap (A - C)$$

Solution:

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8
A	B	C	$B \cup C$	$A - (B \cup C)$	$A - B$	$A - C$	$(A - B) \cap (A - C)$
0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1
0	1	0	1	0	0	0	0
1	1	0	1	0	0	1	0
0	0	1	1	0	0	0	0
1	0	1	1	0	1	0	0
0	1	1	1	0	0	0	0
1	1	1	1	0	0	0	0

It is easy to see Col 5 corresponding to LHS and Col 8 corresponding to RHS are identical.

QUESTION 3. **(33 marks)**

A relation R is defined on set $A = \{a, b, c\}$, as $R = \{(a, a), (a, b), (b, c)\}$.

(a) (24 marks) Find R^{-1}, $R \circ R^{-1}$, $R^{-1} \circ R$, R^2, R^t, and $R \cup R^2$ by listing all elements of the sets.

Solution:

- $R^{-1} = \{(a, a), (b, a), (c, b)\}$
- $R \circ R^{-1} = \{(a, a), (a, b), (b, a), (b, b), (c, c)\}$
- $R^{-1} \circ R = \{(a, a), (b, b)\}$
- $R^2 = \{(a, a), (a, b), (a, c)\}$
- $R^t = \{(a, a), (a, b), (b, c), (a, c)\}$
- $R \cup R^2 = \{(a, a), (a, b), (b, c), (a, c)\}$

(b) Answer the following questions. (You need not justify your answer and the justification will not be graded.)

(i) (3 marks) Is $R \circ R^{-1}$ reflexive ?

(ii) (3 marks) Is $R^{-1} \circ R$ reflexive ?

(iii) (3 marks) Is $R \cup R^2$ transitive ?

Solution:

(i) YES

(ii) NO

(iii) YES

QUESTION 4. **(32 marks)**

Denote $A = \{x \in \mathbb{Z} \mid 1 \leq x \leq 1812 \text{ and } 4|x\}$, and $B = \{x \in \mathbb{Z} \mid 1000 \leq x \leq 2023 \text{ and } 6|x\}$, determine the size of sets A, B, $A \cap B$, and $A \cup B$. **Solution:**

We have $|A| = 1812/4 = 453$.

Since $6 \nmid 1000$, we have $|B| = \lfloor 2023/6 \rfloor - \lfloor 1000/6 \rfloor = 171$.

Observe that $A \cap B = \{x \in \mathbb{Z} | 1000 \leq x \leq 1812 \text{ and } 12|x\}$. Since $12 \nmid 1000$, we have $|A \cap B| = \lfloor 1812/12 \rfloor - \lfloor 1000/12 \rfloor = 68$.

Finally, $|A \cup B| = |A| + |B| - |A \cap B| = 556$.

BONUS QUESTION. **(10 marks)**

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

How many integers in the range $[100000, 999999]$ (inclusive of both 100000 and 999999) contain exactly two unique digits (e.g., 122222, 343434) ?

Solution:

The first digit has 9 choices. Fix the first digit, consider the positions of this digit in the remaining 5 digits. There are $2^5 - 1$ ways to choose such positions (minus 1 comes from excluding putting the digit on all remaining 5 digits). For the rest positions, there are 9 choices to choose a different digit. The answer is therefore

$$9 \cdot 31 \cdot 9 = 2511.$$